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Summary

Machine Learning researcher and Imaging Science PhD candidate specializing in **medical AI, computer vision, self-supervised representation learning, and continual learning**. Experienced in building deep learning systems for **2D and 3D imaging**, including **CT/MRI preprocessing, self-supervised pretraining, segmentation, detection, classification, and model evaluation**. At Siemens Healthineers, developed reproducible **PyTorch/MONAI** training and evaluation workflows for **3D chest CT** and pretrained transformer backbones transferable to **organ classification, organ segmentation, and lung nodule detection**. Published 7+ peer-reviewed papers at **ICML, NeurIPS, TMLR, and BMVC**, with experience developing research prototypes, evaluating model performance, analyzing limitations, and communicating results to technical collaborators.

Technical Skills

- **Languages/Frameworks:** Python, PyTorch, MONAI, SimpleITK, Pandas, Scikit-learn, Azure AI ML.
- **Deep Learning:** CNNs, Transformers, Self-Supervised Learning, Continual Learning, Representation Learning, Feature Extraction, Segmentation, Detection, Classification, Transfer Learning, Multi-GPU Training.
- **Medical AI:** 3D CT, MRI, DICOM Processing, Medical Image Preprocessing, Resampling and Spacing, Intensity Normalization, CT Windowing, 3D Augmentation, Quantitative Image Analysis, Model Evaluation.

Industry Experience

Siemens Healthineers 

Malvern, PA, USA 

ML Research Intern

August 2023 – November 2023

- Designed and implemented **3D self-supervised pretraining** for chest CT, producing a pretrained transformer backbone transferable across **organ classification, organ segmentation, and lung nodule detection**.
- Built reproducible **PyTorch/MONAI training and evaluation pipelines** for 3D CT data, including preprocessing, training, validation, and downstream evaluation.
- Improved downstream iteration efficiency (**up to 1000× fewer iterations**) for a transformer-based 3D lung nodule detection workflow through effective pretraining and representation transfer.
- *Technologies used:* Python, PyTorch, MONAI, Azure AI ML, 3D CT data

NTT DATA Europe & LATAM 

Lima, Peru 

Data Scientist

May 2018 – July 2019

- Owned model training and evaluation for a **recommender system** that personalized promotions for **1.5M clients**; supported production deployment and communicated model performance to stakeholders.
- Developed **CNN computer vision models** for applied visual classification tasks, including vehicle damage severity categorization and product recognition.
- Built a **video analytics pipeline** for indoor space usage estimation using object detection and tracking.
- *Technologies used:* Python, PyTorch, Keras, TensorFlow Lite, OpenCV, Docker, Flask, PySpark, DVC

Siemens Healthineers 

Malvern, PA, USA 

ML Research Intern

April 2017 – March 2018

- Designed and implemented a **3D CNN for lung nodule classification** on chest CT volumes, maintaining **90% sensitivity** while reducing **false positives per patient by 50%** compared with a production baseline.
- Built an annotation and case review tool to support expert label curation, iterative error analysis, and review of challenging chest CT cases.
- Presented model results, failure modes, and workflow usability findings to technical stakeholders.
- *Technologies used:* Python, Caffe, SimpleITK, Tkinter, 3D chest CT data

Universidad Nacional de Ingeniería 

Lima, Peru 

Research Assistant

July 2016 – March 2017

- Developed a **CNN medical image analysis model** for diabetic retinopathy detection from retinal fundus images, achieving 83% accuracy.
- Evaluated model performance and contributed to a publication on medical image classification.
- *Technologies used: Python, Keras, OpenCV*

IBM 

Lima, Peru 

Software Tester (QA)

April 2016 – October 2016

- Executed QA test strategies for a telecom database system; collaborated with engineers to reproduce issues, validate fixes, and improve software reliability.
- Built a prototype computer vision workflow using IBM Watson Visual Recognition services.
- *Technologies used: Python, SQL, OpenCV, Watson Visual Recognition*

Selected Project

MRI Fast Acquisition Enhancement ([Project Link](#))

- Built a complete **MRI image quality enhancement pipeline** to restore knee MRI slices under simulated fast acquisition degradation, using a 2D U-Net.
- Implemented DICOM-to-HDF5 conversion, per-volume normalization, degradation simulation, case-level splits, and evaluation on held-out test cases.
- Evaluated image quality with **PSNR, SSIM, and MAE**, supported by qualitative comparisons and error maps.
- *Technologies used: Python, PyTorch, MONAI, PyTorch Lightning Fabric, DICOM, HDF5*

Education

Rochester Institute of Technology 

Rochester, NY, USA 

Ph.D. Imaging Science

August 2019 – August 2026 (expected)

- **Research focus:** Self-supervised and continual representation learning for computer vision.
- **Relevant coursework:** Deep Learning for Vision, Human Visual System, Image Processing, Computer Vision
- **Highlights:** 7+ publications at ICML, NeurIPS, TMLR, BMVC, and other venues.

Universidad Nacional de Ingeniería 

Lima, Peru 

B.Sc. Mechatronics Engineering

March 2011 – December 2015

- **Relevant coursework:** Machine Learning, Image Processing, Object Oriented Programming
- **Highlights:** 2 publications

Selected Publications

Controlling Neural Collapse Enhances Out-of-Distribution Detection and Transfer Learning 

ICML 

Md.Y. Harun, J. Gallardo, and C. Kanan

2025

What Variables Affect Out-of-Distribution Generalization in Pretrained Models? 

NeurIPS 

Md.Y. Harun, K. Lee, J. Gallardo, G. Krishnan, and C. Kanan

2024

Human Emotion Estimation through Physiological Data with Neural Networks 

SoSE 

J. Gallardo, C. Savur, F. Sahin, and C. Kanan

2024

SIESTA: Efficient Online Continual Learning with Sleep 

TMLR 

Md.Y. Harun*, J. Gallardo*, T.L. Hayes, R. Kemker, and C. Kanan

2023

Self-Supervised Training Enhances Online Continual Learning 

BMVC 

J. Gallardo, T.L. Hayes, and C. Kanan

2021

Detection of Diabetic Retinopathy Based on a CNN Using Retinal Fundus Images 

ICANN 

G. García, J. Gallardo, A. Mauricio, J. López, and C. Del Carpio

2017

* *Equal contribution*

Research Service

Reviewer: WACV 2026; AAAI 2025–2026; ICCV 2025; CoLLAs 2025; CVPR 2024–2025; CLVISION 2024–2025